

Duke Energy support for the development of the AI Action Plan

A reliable and abundant energy supply is the bedrock upon which American artificial intelligence (AI) innovation, deployment and dominance will be built. Duke Energy stands ready to provide the energy needed to unleash AI technology's powerful potential in America. In this era of unparalleled opportunity, we are investing $83 billion to add 12.5 gigawatts of energy by 2030. That's enough to power 10 million homes – twice the number that are in our home state of North Carolina. Ensuring a steady and secure power supply 24/7 is critical to serve the growing energy needs of AI while keeping costs affordable for our customers across the Southeast and Midwest. Smart policy solutions can provide the additional energy generation needed and advance our shared goal of building resilient and modern energy infrastructure to power our growing AI-enabled economy.

Key Actions to Accelerate Speed to Market for New Dispatchable Resources by 2-4 years

- *Streamline federal regulatory approval process*
- *Expedite environmental permits, including by working with states*
- *Increase certainty of natural gas supply pipelines*
- *Accelerate domestic production to improve critical equipment delivery times*
- *Fast track interconnection of power plants to the grid*

How policy can help

1. Promote American-made supplies of critical components to strengthen the energy generation supply chain
2. Expedite siting and permitting for critical energy infrastructure while protecting the environment
3. Fast track grid interconnection for new generation facilities necessary to support AI data centers
4. Focus on grid stability
5. Unlock grid flexibility through data center demand response
6. Unleash private investment in energy infrastructure
7. Enhance energy grid defense

We're moving and leading the industry – Duke Energy fast facts

- Operating ~55 Gigawatts of energy capacity for 8.4 million electric and 1.7 million natural gas customers
- Investing $83B across our seven-state footprint to increase generation capacity by more than 20% (nearly 12.5 gigawatts by 2030)
- Developing new customer solutions to meet evolving needs while protecting reliability and affordability for all customers
- Protecting against cyberattacks and natural disasters by meeting high standards for grid security and resilience
- Partnering with the federal government, U.S. military and technology companies to provide generation and secure grid access to critical installations
- Helping to finance the next generation of nuclear by earning more than half a billion of nuclear production credits in 2024, which flow dollar for dollar back to customers

DUKE ENERGY

BUILDING A **SMARTER** ENERGY FUTURE ®

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Comments from Duke Energy
in Response to the Networking and Information Technology Research and Development National Coordination Office's Request for Information on the Development of an Artificial Intelligence Action Plan
March 15, 2025[1]

Executive Summary

Duke Energy Corporation ("Duke Energy" or the "Company") appreciates the opportunity to submit comments to the United States Office of Science and Technology Policy ("OSTP") to support President Trump's direction for his administration to develop the Artificial Intelligence ("AI") Action Plan to advance America's AI dominance.[2] Recognizing that reliable and abundant energy is the bedrock upon which American AI innovation, deployment, and dominance will be built, Duke Energy outlines seven energy-related priorities that will help the United States meet its new, AI-driven data center electric load demands, thereby enabling the country to enhance its economic strength, buttress its national security objectives, and support the American people.

Duke Energy provides these comments based on its over 100-year history of providing the energy that has driven economic growth and prosperity for the regions and people it serves. As a Fortune 150 company headquartered in Charlotte, N.C., Duke Energy is one of America's largest energy holding companies. The Company's electric utilities serve 8.4 million customers in North

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[1] Comments submitted electronically to [redacted] Office of Science and Technology Policy.

[2] On January 23, 2025, President Trump signed Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence), which declared that it is the policy of the United States to sustain and enhance America's global AI dominance to promote human flourishing, economic competitiveness, and national security. Executive Order 14179 also directed the development of an AI Action Plan ("AI Plan") designed to achieve policy goals. On February 6, 2025, the Networking and Information Technology Research and Development National Coordination Office published a Request for Information on behalf of the OSTP requesting input from the public, including private sector organizations, on priority actions that should be included in the AI Plan.

Carolina, South Carolina, Florida, Indiana, Ohio and Kentucky, and collectively own approximately 54,800 megawatts of energy capacity. Its natural gas utilities serve 1.7 million customers in North Carolina, South Carolina, Tennessee, Ohio and Kentucky.

Working with our datacenter/AI customers on innovative partnerships, state legislatures and regulatory commissions on AI and growth supportive policy, supply chain partners and community and technical colleges on workforce development (just to name a few), Duke Energy is advancing on multiple fronts to rise to the call around AI dominance.

Duke Energy is working on innovative, tailored solutions to meet growing, large-scale energy needs in its service areas. For example, to help encourage critical investments necessary for new large loads while protecting existing customers, our contracts include enhanced performance and credit provisions such as security for interconnection costs, elevated minimum bills, and required long-term contracts with termination penalties – all of which help advance critical infrastructure upgrades while reducing risk if loads fail to materialize as expected. This work is aligned with the Trump Administration's energy priorities, as well as its priority to retain America's place as a global leader in AI technology and development. Duke Energy's efforts include meeting the high standards for grid security and resilience, as well as protecting against both physical and cyberattacks and natural disasters.

Duke Energy's customer-centric approach features programs that are evolving and maturing to match the new dynamic needs of our customers and the grid. Duke Energy announced signed customer agreements in the third quarter of last year to connect two gigawatts of new data centers and anticipates load growth to begin accelerating. The Company is ready, willing, and able to power these data centers and meet this unprecedented load growth as a long-standing provider of reliable electric services to high load factor customers across many industries. As a regulated utility, Duke Energy ensures that the Company's actions both add value for all customers connected to the grid while ensuring safe, affordable and reliable energy.

Duke Energy not only has a proven record of planning, developing, and operating energy infrastructure in a safe and reliable manner, but also one of partnering with the federal government, U.S. military, technology companies, and other large customers to meet customers' evolving needs and achieving mutually beneficial outcomes. Indeed, dating back to 1952 when the Company joined ownership of the Ohio Valley Electric Cooperative with the federal government and other investor-owned utilities in the Ohio River Valley area to support the large electric load needed for uranium enrichment facilities then under construction by the Atomic Energy Commission, Duke Energy has supported initiatives that helped implement the vital policy goals of the United States and the federal government. Furthermore, Duke Energy serves significant military installations with peak demand exceeding 200 megawatts, including meeting their specific energy security needs with on-site generation (e.g., innovative microgrid solution) and secure grid access.

Given this demonstrated track record of supporting national priorities, the Company is driven to meet the nation's most pressing challenges, including the rapid deployment of energy solutions for the AI sector.

Sustaining and enhancing the United States' global AI dominance will require meeting the extraordinary growth in energy demand driven in good measure by AI. To meet the President's goals, the AI Plan should also include economic development initiatives that encourage and streamline private investment in energy infrastructure that can support AI.

Duke Energy supports President Trump's goal of meeting AI's massive energy requirements while maintaining reliability and affordability for customers. Specifically, Duke Energy is focused on accelerating the deployment of new electric generation capacity and related infrastructure, ensuring grid stability, and delivering cost-effective, modern, and resilient energy solutions to power the AI revolution. By 2030, Duke Energy plans to increase our generation capacity by nearly 12.5 gigawatts – enough to power approximately 10 million average homes. This increase is equivalent to adding more than 20% of our current generating fleet capacity in the next five years.

Expediting Critical Energy Infrastructure is Essential to AI Leadership

Duke Energy applauds President Trump's recognition that energy dominance and affordability are fundamental not only to America's AI leadership, but vital for promoting national security, good paying jobs, and the quality of life for the American people. Recently signed Executive Orders ("EO"), including EO 14156 (Declaring a National Energy Emergency) ("EO DNEE")[3] and 14154 (Unleashing American Energy) ("EO UAE"),[4] together with EO 14179, highlight the President's commitment to securing America's energy future. In declaring a national energy emergency, EO DNEE identifies energy "transportation" and "generation capacity" as "inadequate," and further identifies "inadequate energy supply and infrastructure," and "insufficient energy production . . . and generation," as the basis of the energy emergency.[5] These three EOs underscore the President's understanding that energy dominance, modern energy infrastructure, energy security, and America's AI leadership are interconnected. The EOs also establish the predicate for triggering specific emergency authorities found in federal statutes. Below, Duke Energy outlines seven priority areas that can help support and expedite the development of energy infrastructure necessary to meet the energy demands needed to achieve the President's goal of American AI dominance, thereby enabling the country to promote its economic growth and prosperity strengthen national security.

Policy Recommendations for Enabling Energy Abundance to Foster American AI Dominance

1. <u>Promote American-Made Supplies of Critical Components to Strengthen the Energy Generation Supply Chain</u>

Duke Energy has complete faith in our domestic suppliers and their responsiveness to this growing market. However, policy can further support robust and resilient supply chains, including domestic manufacturing. The AI Plan should consider supporting supplies and services necessary to build power generation facilities and associated infrastructure to support AI-driven data centers developments. Current lead times for key components in the energy supply chain can range from two to three years (e.g., 24 months for power transformers 48 months for high voltage circuit breakers, and 36 months for gas turbines). There are growing demands for steel, aluminum, and

[3] EO No. 14156, 90 Fed. Reg. 8433 (Jan. 29, 2025).
[4] EO No. 14154, 90 Fed. Reg. 8353 (Jan. 29, 2025).
[5] EO DNEE § 1.

many additional components that are needed to construct power plants, transmission lines, natural gas pipelines to support natural gas generation facilities, and grid operating systems. The availability of skilled workers, such as welders, electricians and engineers, is also constrained.

Delays or the inability to obtain supplies and services could severely challenge the ability of the energy sector to meet AI's projected power demands. Increasing project certainty through permitting and other reforms will send strong signals to suppliers. Supporting access to supplies and services necessary to build power generation facilities and associated infrastructure would also help address this need and help reduce risks relating to construction of new generating resources.

The Defense Production Act ("DPA")[6] is a possible policy tool to assist energy developers gain access to the supplies and services needed to meet AI's energy demands in a timely manner. The DPA provides broad authority for the President to prioritize the performance of contracts and allocation of supplies to advance such actions as the building of generation facilities and natural gas pipelines (e.g. steel, generators, etc.). Specifically, § 101(c) of the DPA permits the President to order priority performance of contracts or allocation of supplies "to maximize domestic energy supplies" and make determinations that such "materials, services, and facilities" are "scarce, critical, and essential . . . to construct or maintain energy facilities[.]"[7] Therefore, the AI Plan should consider using the DPA to increase energy developers' access to supplies and services necessary to build power generation facilities and the construction of associated infrastructure and to support the goal of "build it in America."

2. <u>Expedite Siting and Permitting for Critical Energy Infrastructure While Protecting the Environment</u>

The Company believes Americans can have abundant, modern, affordable, reliable, and resilient energy in support of AI dominance, while also ensuring a healthy environment with clean air and clean water and protecting endangered species. To achieve these goals, however, efficient, expedited and certain environmental permitting is essential. Recognizing these challenges, the

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[6] 50 U.S.C. § 4501 *et seq.*
[7] 50 U.S.C. § 4511(c).

President's EOs DNEE and UAE direct federal agencies to streamline environmental permitting and to use existing authorities to expedite the construction of energy infrastructure.[8] As the AI Plan is developed, a prudent way to promote energy infrastructure development while also protecting the environment is to apply a risk-informed balancing approach. Too often, federal regulators engage in a "no-risk" approach to reviewing permit applications, which is not necessarily consistent with law. Instead, federal agencies should be encouraged to make sound, informed decisions by measuring the likelihood of the risk and the costs to mitigate such risk.

Timely permit issuance through final agency approvals is also essential to meet this moment. Whether it involves building a new power generation facility, an electric transmission line, or a natural gas pipeline to support power generation, obtaining the necessary permits expeditiously, and with certainty, under, among many other statutes, the Clean Water Act ("CWA"),[9] the Clean Air Act ("CAA"),[10] and the Endangered Species Act ("ESA"),[11] is unreasonably difficult. The delays may be caused by interagency disputes and lack of proper coordination or legal challenges against the issuing agencies.

In order to facilitate the execution of projects that support domestic AI development, the Company appreciates EO DNEE's direction to agencies to use existing emergency authorities to grant clean water permits and to require administrative consideration to employ potential nationwide permits under the CWA and other statutes administered by the Army Corps of Engineers.[12] The Company also appreciates EO DNEE's acknowledgement of the need to expedite ESA permits through the so-called "God Squad," while also protecting endangered species.[13]

The process involved to conduct and then finalize an environmental review, led by a federal agency, potentially with several "cooperating agencies," under the National Environmental Policy

[8] EO DNEE §§ 2, 3, 4, 5, and 6; EO UAE §§ 3, 5, and 6.
[9] 33 U.S.C. § 1251 *et seq.*
[10] 42 U.S.C. § 7401 *et seq.*
[11] 16 U.S.C. § 1531 *et seq.*
[12] EO DNEE § 4.
[13] EO DNEE § 5 and 6. It is noteworthy that the ESA already contains provisions for exemptions based on "undue economic hardship," 16 U.S.C. § 1539.

Act ("NEPA")[14] can take years, which does not include additional time to sort through seemingly inevitable legal challenges. We appreciate the White House Council on Environmental Quality's recent memorandum to federal agencies to "revise or establish their NEPA implementing procedures (or establish such procedures if they do not yet have any) to expedite permitting approvals...." Nonetheless, due to recent court decisions and pending legal cases, there remains uncertainty as to how the agencies will manage NEPA reviews and until resolved, could mean further delays.[15] Therefore, the Company urges the Administration to develop additional guidance for agency compliance with NEPA.

EO UAE's directive to facilitate energy "production" on federal lands could also facilitate accelerated deployment of generation facilities.[16] Although state regulatory commissions typically approve certificates for new utility generation facilities on non-federal land, EO DNEE directs the executive branch to exercise its "siting" authorities with respect to "federal lands" and to access the use of federal eminent domain and submit recommendations to the President.[17] Duke Energy has a strong track record and history of serving federal facilities throughout our jurisdictions.

It is likely some of the new generating facilities that will power AI across the nation will be powered by natural gas, which means there will be a need for additional natural gas pipelines to deliver the gas to those facilities. Federal authorizations for interstate natural gas pipelines fall within the Natural Gas Act ("NGA")[18] and are administered by the Federal Energy Regulatory Commission ("FERC"). Given that § 7 of the NGA already provides eminent domain authority for holders of NGA certificates,[19] the AI Plan should consider policies that would support and facilitate the targeted and reasonable use of authorities to construct necessary new and incremental

[14] 42 U.S.C. § 4321 *et seq.*
[15] The NEPA process is in doubt because of the recent decision by in *Marin Audubon Society v. Federal Aviation Administration*, No. 23-1067 (D.C. Cir. Nov. 12, 2024); den. *en banc*., where the court held that the Council on Environmental Quality ("CEQ") did not have the authority to issue NEPA regulations. President Trump's EO UAE directs the CEQ to propose rescinding the NEPA regulations.
[16] EO UAE § 2; though not defined in EO UAE, "production" is defined in EO DNEE § 8.
[17] EO DNEE § 2.
[18] 15 U.S.C. § 717 *et seq.*
[19] 15 U.S.C. § 717f.

facilities that would support the development of natural gas-fired dispatchable generation to support AI-driven data center development. Another potential option for further consideration is that, pursuant to the Natural Gas Policy Act ("NGPA"),[20] the President can declare a natural gas supply emergency and may issue an order to "*require* any pipeline to transport natural gas, *and to construct and operate such facilities for the transportation of natural gas*, as he determines necessary" to carry out executive rules or orders in place during a natural gas supply emergency.[21] These are initial examples of statutory authority granted to the President to ensure the construction and operation of natural gas facilities under national emergency circumstances.

Therefore, the Company supports the AI Plan in identifying ways to expedite environmental permitting and siting for critical energy infrastructure to support AI, while also protecting the environment.

3. Fast Track Grid Interconnection for New Generation Accompanying Large Loads

To enable sufficient generation production to meet the electricity demand of AI technologies, the AI Plan should include a process to fast track the interconnection of new generation facilities necessary to support AI data centers. As is well documented, the interconnection queues for new generation are combined with projects that may or may not be built and may not be the type of resources needed to meet the energy demands and load profiles of AI data centers.

Expedited interconnection can be accomplished through § 202(c) of the Federal Power Act ("FPA"), which states that the U.S. Department of Energy ("USDOE") can order the "temporary" connection of facilities for the generation and transmission of electricity to meet the "public interest."[22] Such orders that "may result in a conflict with a requirement of any Federal, State, or local environmental law or regulation" must be renewed every 90 days. FPA § 202(c) is an

[20] 15 U.S.C. § 3301 *et seq*.
[21] 15 USC § 3362(c) (emphasis added).
[22] 16 U.S.C. § 824a(c). *See also*, 10 C.F.R. Part 205, § 205.371 Definition of emergency.

effective tool with which the industry is already familiar, as USDOE regularly exercises this authority.

4. Focus on Grid Stability

The AI Plan should include a directive for AI data center developers to work with generation, transmission, and distribution asset owners to ensure that (1) new data centers are built in locations that do not overburden the electric grid, and (2) new grid infrastructure necessary to maintain the reliability of the system is identified and constructed along with the respective AI data center. Our Company has been strongly interlinked with the communities we serve for many years, including more than a century in some locations. We work closely with them to understand and support their changing needs and have a long history of serving communities with excellence and well-established infrastructure to ensure reliable, 24/7 energy service. This directive should also consider incentivizing AI data center construction in areas with existing generation resources and existing infrastructure so that utilities can leverage previous facility investments to support AI development. Such a directive would help to ensure the availability of reliable and economic power for both AI technologies and other types of consumers.

5. Unlock Grid Flexibility Through Data Center Demand Response

The AI Plan should consider incentives for the development and deployment of energy-efficient AI data center technologies, such as advanced cooling systems and intelligent power management, as well as other strategies for enhancing data center flexibility. Development and deployment of such technology would help lessen the demands that AI development and implementation will impose on the nation's energy infrastructure which will, in turn, facilitate its successful integration.

The AI plan should seek to allow additional pathways for flexible load response so as to alleviate potential shortages during peak demand periods on the grid. Supporting the expansion of demand-side flexibility through technological advancements in battery storage, nuclear generation and renewables, along with expanded fuel options for backup facilities, could help accelerate the onboarding of AI data centers, while also ensuring reliability for all customers. Supporting the expansion of demand-side flexibility through technological advancements in battery storage, nuclear generation and renewables, along with expanded fuel options for backup facilities, will undoubtedly accelerate the onboarding of AI data centers, while also ensuring reliability for all customers.

6. Unleash Private Investment in Energy Infrastructure

The AI Plan should implement economic development drivers that encourage and streamline private investment in energy infrastructure. Investments in energy infrastructure are undoubtedly capital-intensive endeavors that will serve customers and communities for decades. Financing this critical infrastructure requires predictable cost recovery frameworks, including tax policy, which maintain customer affordability while providing the stability upon which utility sector investors have grown accustomed to depending. Such tax policies not only help support investment in capital intensive infrastructure, but they can also help mitigate impacts on customers. For example, Duke Energy customers will receive – dollar-for-dollar - more than half a billion dollars in nuclear production credits in 2024, which flow back to customers quickly through an important accelerator for nuclear projects called transferability. This pairing of the nuclear PTC and transferability is essential to finance the next generation of nuclear energy.

The option of new full-scale or small modular nuclear power development requires significant upfront capital, with long development and regulatory timelines. In addition to exploring ways to streamline the regulatory process, the federal government should also explore facilitating financial mechanisms for both full-scale and small modular reactors, such as loan guarantees, cost overrun insurance for early movers, reinforce long-term market structures, and overall improve the economic viability of new nuclear projects and attract private investment. Over time, policies like these would help nuclear power transition from reliance on federal support to a self-sustaining, market-driven energy source that can provide reliable, carbon-free electricity for AI data centers and the broader grid.

7. Energy Grid Defense

The United States' electric grid is a critical national security asset upon which enhancement of the United States' position as a global leader will depend. Accordingly, the AI Plan should deploy AI defense systems to protect the electric grid from cyber and physical attacks at a level commensurate with the electric grid's integral importance to the United States's national security. The AI defense system requirements and solutions should be established in tight partnership and collaboration with the critical infrastructure providers to ensure protection of the electric grid without compromising utilities' ability to effectively secure the electric grid.

DPA § 106 designates "energy "as a strategic and critical material,[23] and DPA § 702's definition of "national defense" includes programs for "energy production and construction."[24] Therefore, the AI Plan should consider the deployment of the full scope of AI capabilities to protect, to the greatest extent possible, the electric grid from adversaries that would seek to harm the United States. Of course, this also means such AI will need reliable and secure energy sources.

Conclusion

The United States' ability to achieve AI dominance is inextricably linked to its energy infrastructure. Duke Energy is committed to providing the reliable, affordable, and abundant energy necessary to power America's AI revolution.

Duke Energy thanks the President Trump for his leadership on this important issue and the Company thanks his Administration for providing the opportunity to submit input on the development of an AI Action Plan. The Company looks forward to partnering with the Trump Administration with respect to the successful implementation of the EOs referenced in these comments, including EO 14179.

[23] 50 U.S.C. § 4516.
[24] 50 U.S.C. § 4552(14).